

Quantum Semantic Communications

Context and motivation.

Future communication systems are expected to transmit not only bits, but information that is useful and meaningful to the receiver. Semantic communication addresses this challenge by transmitting a compact representation of the meaning or task-relevant content of a message rather than reproducing it bit by bit. Recent work has started to combine this idea with quantum communication, leading to the emerging field of **Quantum Semantic Communications (QSC)**.

The objective of this project is to investigate approaches in which quantum representations, quantum correlations, superposition, contextuality or quantum geometry contribute directly to the modelling and transmission of semantic information. The project will build on recent QSC proposals while connecting them with quantum semantics and quantum-based models of textual meaning.

Methodology.

First, the students will establish a structured state of the art covering QSC, quantum semantics, semantic communication and quantum information theory.

The students will then implement a small demonstrator, for example a quantum representation of words/concepts or a graph-based semantic representation, followed by transmission through simulated depolarizing, dephasing and amplitude-damping channels.

Some references

1. M. Chehimi, C. Chaccour, C. Kurisummoottil Thomas, W. Saad, *Quantum semantic communications for resource-efficient quantum networking*, 2024.
2. M. Wang, G.-F. Zhu, G.-F. Long, J. Guo, Y.-C. Liu, D. Pan, L.-P. Nong, G.-L. Long, *Quantum semantic communication beyond the Shannon-Wyner channel capacity*, 2025.
3. N. Nunavath, M. I. Habibie, E. Calvanese Strinati, R. Bassoli, F. H. P. Fitzek, *Quantum semantic communications for graph-based models*, IEEE SPAWC, 2024.
4. F. Galofaro, Z. Toffano, B.-L. Doan, *A quantum-based semiotic model for textual semantics*, Kybernetes, 2018.
5. I. A. Surov et al., *Quantum semantics of text perception*, Scientific Reports, 11, 4193, 2021.
6. F. Dubois, Z. Toffano, *On Quantum Relations*, Entropy, 28, 522, 2026.

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